mmFHE: mmWave Sensing with End-to-End Fully Homomorphic Encryption

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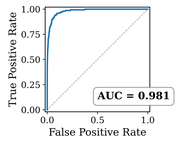
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Raw mmWave Data Reveals Too Much

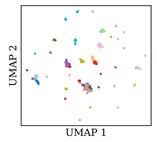
mmWave radar enables wireless **heart rate, breathing, sleep, gait, and gestures** sensing. But offloading raw data to the cloud computation creates two problems:

1 Biometric leakage Prop. 2.1
The raw stream carries identity, heartbeat, and breathing, so the cloud can re-identify and link users.

2 DSP leakage Prop. 2.2
Standard data-dependent DSP (CFAR, argmax, tracking) reveal the scene to the cloud.



(a) Linkage of 50 children, no training



(b) 12 users cluster without labels

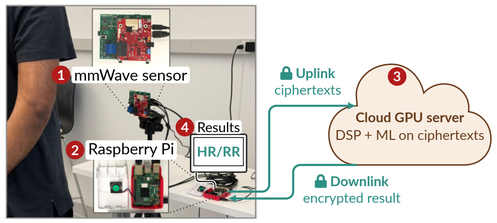
99.9% user re-ID within a session (12 users)

78.6% re-ID across different gesture types

0.981 linkage AUC with no labels or training

mmFHE Enables Sensing on Encrypted Data

mmFHE enables mmWave sensing on **encrypted data** using fully homomorphic encryption (FHE). The cloud runs all its **DSP and ML** on ciphertexts, so it does the sensing without ever seeing the raw data.



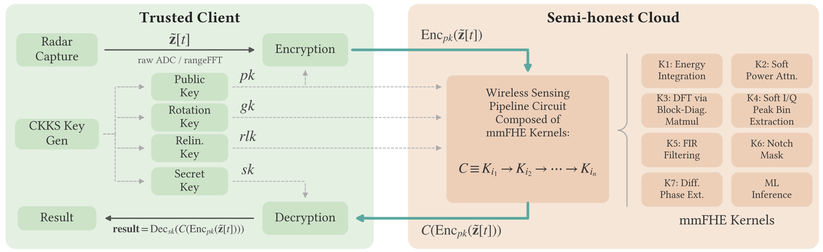
FHE Primer

- **FHE** allows only three *primitive operations* on ciphertexts: **add**, **multiply**, **rotate**.
- **CKKS** scheme extends FHE over approximate real and complex numbers.
- **Depth budget:** at most L multiplications in a row before costly bootstrapping.

Threat Model

- **Adversary:** a *semi-honest* cloud. It runs the protocol faithfully but analyzes every input, intermediate value, and timing. It cannot break RLWE.
- **Trust boundary:** the edge client is *trusted*. It stores the secret key, encrypts, and decrypts. The cloud is *semi-honest*, gets only ciphertexts and public keys.
- **Goals:**
  - **Input privacy** Thm 4.1 — The cloud learns nothing about the encrypted sensor stream.
  - **Data obliviousness** Thm 4.2 — The execution trace is the same no matter what is sensed.
- **Assumptions:** malicious cloud deviating from protocol, network metadata, session timing are out of scope.

mmFHE System Design



Pre-Cloud Client-Side Algorithms

1. **Preprocess:** remove static clutter, then scale to a stable range.
2. **Encrypt:** CKKS, with real and imaginary parts in separate ciphertexts.
3. **Batched packing:** (i) single frame, or (ii) multiple frames are "batched" in each ciphertext.

Cloud-Side Algorithms

mmFHE kernel	Standard step	Enables
K1 **Energy Integration**	Energy accumulation	per-bin energy over the window
K2 **Soft Power Attention**	CFAR / argmax (branches)	branch-free target-bin selection
K3 **DFT**	FFT (1 level per stage)	frequency domain analysis
K4 **Soft I/Q Extraction**	Peak-bin I/Q (argmax)	target I/Q with no hard selection
K5 **FIR Filtering**	FIR band-pass	respiration and heart-rate bands
K6 **Notch Masking**	Static-clutter notch	clutter removal with a fixed mask
K7 **Differential Phase Extraction**	atan2 + unwrap (division)	phase change, no arctan or division
ML **Encrypted ML Inference**	ReLU (non-polynomial)	classification with x^2 activations

Evaluation Setup

- **Data** (i) **Children**, (ii) **Parralejo** (50 and 110 subjects): 270 vital-sign recordings (iii) **60 GHz Gesture** (12 users, 5 classes): 600 gesture recordings
- **Baseline** the identical pipeline run in plaintext, and RPRS
- **Hardware** OpenFHE + FIDESlib on single RTX 3090 Ti GPU; Raspberry Pi 4 client

Example Pipeline for Vital Sign Monitoring

Heart-Rate and Breathing Detection using mmWave. HR / RR (waveform or BPM)

K1 → K2 → K4 → K5 → K7 → K3 → K1 → K2

mmFHE vs. plaintext baseline pipeline, not clinical accuracy (†lying, ‡sitting)

Dataset	n	HR MSE ↓	RR MSE ↓	HR/RR rate MAE ↓
Children	50	0.264	0.181	$<10^{-3}$ bpm
Parralejo	$110^{\dagger}/110^{\ddagger}$	$0.064^{\dagger}/0.136^{\ddagger}$	$0.074^{\dagger}/0.179^{\ddagger}$	$<10^{-3}$ bpm

mmFHE introduces negligible error in vital-sign detection

Example Pipeline for Gesture Classification

5-way Gesture Detection using mmWave. logits

K3 → K1 → K6 → K2 → Σ_t → FC·x^2 → FC·x^2 → FC

Method	Encrypted	Accuracy	Latency
Plaintext	nothing	95.0%	0.04 ms
RPRS	CNN only	95.0%	14.09 s
mmFHE (ours)	**DSP + CNN**	84.5%	**5.76 s**

The drop is from kernel approximation, not encryption (vs. plaintext: 84.7%).

mmFHE is the first system to implement both encrypted DSP and ML for mmWave sensing

FHE Parameters and Encryption Error

FHE Parameter	mmFHE
FHE scheme	CKKS
Library	OpenFHE + FIDESlib (GPU)
Polynomial degree	32,768
Slots	16,384
Encrypted stages	DSP + CNN
Multiplicative depth	11
Security level	128-bit classic

Kernel	MSE	Max |err|	Depth
K1	4.1×10^{-14}	1.5×10^{-6}	1
K2	5.5×10^{-6}	6.9×10^{-3}	2
K3	4.8×10^{-22}	7.8×10^{-11}	1
K4	4.6×10^{-14}	1.0×10^{-6}	3
K5	1.5×10^{-14}	5.2×10^{-7}	4
K6	8.2×10^{-24}	7.0×10^{-12}	3
K7	5.3×10^{-15}	2.3×10^{-7}	5

Computational Overhead

Per window (per-frame → batched)	Vital signs	Gesture
Input ciphertexts	800 → 4	200 → 26
Uplink	4.3 GB → 22 MB	1.2 GB → 156 MB
Downlink	33 MB → 33 MB	6 MB → 6 MB
Key material (one-time)	3.1 GB → 1.4 GB	4.4 GB → 2.7 GB
Client encrypt (RPi 4)	164 s → 0.84 s	44.3 s → 5.9 s
Cloud GPU memory (peak)	20.7 GB → 5.3 GB	4.3 GB → 7.5 GB

mmFHE works on COTS edge (Raspberry Pi 4) and GPU (RTX 3090 Ti)

Latency

Pipeline	Plaintext	mmFHE per-frame	mmFHE batched
Vital signs (10 s window)	2.79 ms	99.48 s (35,656×)	**1.21 s** (434×)
Gesture (3 s window)	13.55 ms	36.56 s (2,699×)	**5.76 s** (425×)

10 s vital window takes 1.21 s and 3 s gesture window takes 5.76 s

Limitations and Future Work

- **Limitations:** semi-honest cloud, one line-of-sight target, batch only.
- **Future work:** multi-target, streaming, and verifiable FHE.
- **Acknowledgment:** Digital Life Initiative Doctoral Fellowship, Cornell Tech.